

MCQ-8

Topic: Machine Learning – Classification, Trees, kNN, SVM & Evaluation Metrics

Student Name: _____ **Date:** _____ **Score:** _____

Instructions: Select the correct option by marking the checkbox ☐ next to it.

1. Suppose $P(\text{Spam}) = 0.4$ and $P(\text{NotSpam}) = 0.6$. For an email containing word W , $P(W \mid \text{Spam}) = 0.75$ and $P(W \mid \text{NotSpam}) = 0.20$. What is $P(\text{Spam} \mid W)$?
☐ 0.7500
☐ 0.8333
☐ 0.7143
☐ 0.8824
2. Why can an unrestricted decision tree overfit the training data?
☐ Because decision trees require normally distributed features.
☐ Because each split necessarily increases the test-set error.
☐ Because decision trees cannot represent nonlinear decision boundaries.
☐ Because the tree can continue splitting until leaves contain very small or pure subsets.
3. A logistic regression model has $z = -1.2 + 0.8x_1 + 0.5x_2$. For an observation $x_1 = 2, x_2 = 1$, what is the predicted probability of the positive class?
☐ 0.7109
☐ 0.6900
☐ 0.7685
☐ 0.8320
4. Suppose a kNN classifier uses $k = 1$ instead of $k = 15$. Which behavior is most likely?
☐ Higher variance and lower bias
☐ Lower variance and higher bias
☐ Higher bias and higher variance
☐ Lower bias and lower variance
5. A node contains 100 observations: 70 belong to Class A and 30 belong to Class B. Using base-2 logarithms, what is the approximate entropy?
☐ 0.610
☐ 0.910
☐ 0.970
☐ 0.881
6. A binary classifier produces $TP = 72$, $TN = 108$, $FP = 18$, and $FN = 12$. What is the precision of the classifier?

- ☐ 0.8571
- ☐ 0.8000
- ☐ 0.8889
- ☐ 0.9000

7. In a linearly separable SVM, the support vectors are:

- ☐ The observations closest to the decision boundary that determine the separating hyperplane.
- ☐ All observations that lie on the correct side of the decision boundary.
- ☐ The observations incorrectly classified by the final model.
- ☐ The observations with the largest feature values.

8. A decision-tree node contains 40 samples: 24 belong to Class A and 16 belong to Class B. What is the Gini impurity of the node?

- ☐ 0.40
- ☐ 0.48
- ☐ 0.52
- ☐ 0.60

9. Which statement about logistic regression is most accurate?

- ☐ It always produces perfectly calibrated probabilities regardless of the dataset.
- ☐ It directly predicts a class label by minimizing squared classification error.
- ☐ It models the log-odds of the positive class as a linear function of the input features.
- ☐ It assumes that every input feature follows a Gaussian distribution.

10. Consider the dataset below:

Point	Coordinates	Class
A	(1, 1)	Red
B	(2, 2)	Red
C	(3, 1)	Blue
D	(4, 2)	Blue
E	(2, 3)	Red

For query point $Q = (3, 2)$, using $k = 3$ and Euclidean distance, what is the predicted class?

- ☐ Blue, because 2 of the 3 nearest neighbors are Blue
- ☐ Red, because 2 of the 3 nearest neighbors are Red
- ☐ Red, because the total number of Red points is larger
- ☐ Blue, because the average distance to Blue points is smaller

11. A parent node has entropy 0.970. After splitting on feature X, the left node contains 60 samples with entropy = 0.722, and the right node contains 40 samples with entropy = 0.811. What is the approximate information gain?

- ☐ 0.212
- ☐ 0.183
- ☐ 0.243
- ☐ 0.287
12. Using $TP = 72$, $TN = 108$, $FP = 18$, and $FN = 12$, what is the approximate F_1 -score?
- ☐ 0.8182
- ☐ 0.8571
- ☐ 0.8276
- ☐ 0.8889
13. The “naive” assumption in Naive Bayes refers primarily to:
- ☐ Assuming the features are conditionally independent given the class.
- ☐ Assuming all classes have equal prior probability.
- ☐ Assuming all features have the same probability distribution.
- ☐ Assuming the training data contains no missing values.
14. Consider a linear SVM with parameters $W = [2, -1]$ and $b = -3$. For point $x = [2, 1]$, calculate $f(x) = w^T x + b$. What is the result?
- ☐ -1
- ☐ 0
- ☐ -2
- ☐ +1
15. Why is feature scaling particularly important for kNN?
- ☐ Scaling prevents kNN from overfitting by reducing the number of training samples.
- ☐ Scaling converts categorical variables into continuous variables automatically.
- ☐ Scaling ensures that all features contribute equally to the distance calculation.
- ☐ Scaling guarantees that the nearest neighbor belongs to the correct class.
16. A decision-tree algorithm evaluates two possible features: Feature A produces an Information Gain of 0.31, and Feature B produces an Information Gain of 0.24. Assuming all other conditions are identical, which feature should normally be selected?
- ☐ Feature B, because smaller information gain indicates less overfitting.
- ☐ Feature A, because it produces the larger reduction in entropy.
- ☐ Feature B, because information gain should be minimized.
- ☐ Either feature, because information gain does not influence split selection.
17. A medical screening system is designed to detect a rare disease. Missing an actual positive case is significantly more costly than incorrectly identifying a healthy person as positive. Which metric should generally receive the greatest emphasis?
- ☐ Precision
- ☐ Recall
- ☐ Specificity
- ☐ Negative Predictive Value

18. A parent node contains 100 samples (60 Class A, 40 Class B). A candidate split produces a left child with 50 samples (40 A, 10 B) and a right child with 50 samples (20 A, 30 B). What is the weighted Gini impurity after the split?
- ☐ 0.40
 - ☐ 0.44
 - ☐ 0.46
 - ☐ 0.48
19. A dataset contains two classes arranged approximately as concentric circles. A linear decision boundary performs poorly. Which approach is most appropriate for an SVM?
- ☐ Increase the learning rate while keeping a linear kernel.
 - ☐ Remove the support vectors before training.
 - ☐ Use a nonlinear kernel such as the RBF kernel.
 - ☐ Increase the number of classes to make the boundary nonlinear.
20. Which statement correctly compares Gini impurity and entropy?
- ☐ Both measure node impurity, but entropy uses logarithms while Gini does not.
 - ☐ Gini can only be used for binary classification, while entropy supports multiclass classification.
 - ☐ Entropy always produces deeper trees than Gini.
 - ☐ Gini measures information gain directly, while entropy measures classification accuracy.